SonicMAE: Complete Project Workflow & Technical Architecture

Domain: Industrial Acoustic Condition Monitoring & Unsupervised Anomalous Sound Detection
Benchmark Dataset: MIMII DUE Industrial Water Pump Challenge
GitHub Repository: https://github.com/Amrutha-Tsavatapalli/SONIC_MAE
Document Status: Complete Technical Workflow Specification

1. Executive Summary & Problem Formulation

Industrial centrifugal water pumps operating in manufacturing and municipal water facilities deteriorate over time due to bearing raceway flaking, blade cavitation, and seal wear. Detecting early-stage anomalies before catastrophic breakdown is vital to reduce unplanned downtime and avoid safety hazards.

The Fundamental Constraint: Extreme Class Imbalance. In real-world factories, nominal running sounds are abundant, but failure recordings are scarce, unpredictable, and dangerous to collect. Traditional supervised classifiers cannot be trained.

SonicMAE provides an unsupervised deep self-supervised framework. It trains strictly on normal running audio, reconstructs masked log-Mel spectrogram patches (25% ratio), enforces machine invariance using **Physics-Preserving Conjugate-Symmetric Phase Scrambling (CSPS)**, and detects anomalies using dual-stream scoring with multi-centroid prototype memory.

2. End-to-End System Workflow Architecture



3. Audio Signal Processing & Mathematical Formulation

3.1 Log-Mel Spectrogram Extraction

Audio is peak-normalized: x_norm = x / max(|x|, 1e-8). The Short-Time Fourier Transform (STFT) is calculated with window N=1024 (64 ms at 16 kHz) and hop length H=512 (32 ms, 50% overlap). A 64-band triangular Mel filterbank matrix projects linear frequency to perceptual acoustic bands, compressed logarithmically: S = log(Mel_power + 1e-6). Spectrograms are standardized to 320 frames (10 seconds), yielding tensor shape (1, 1, 64, 320).

3.2 Conjugate-Symmetric Phase Scrambling (CSPS) — Core Novelty

Standard audio data augmentations (pitch shifting, speed changes) alter fundamental rotational speed (f_0) and motor harmonics (k * f_0), violating physical machinery dynamics.

Mathematical Invariance Proof of CSPS

Let X[k] = |X[k]| * exp(j * angle(X[k])) be the complex Fourier spectrum. CSPS adds bounded phase perturbation delta_theta ~ Uniform(-pi * lambda, pi * lambda) with lambda = 0.20:

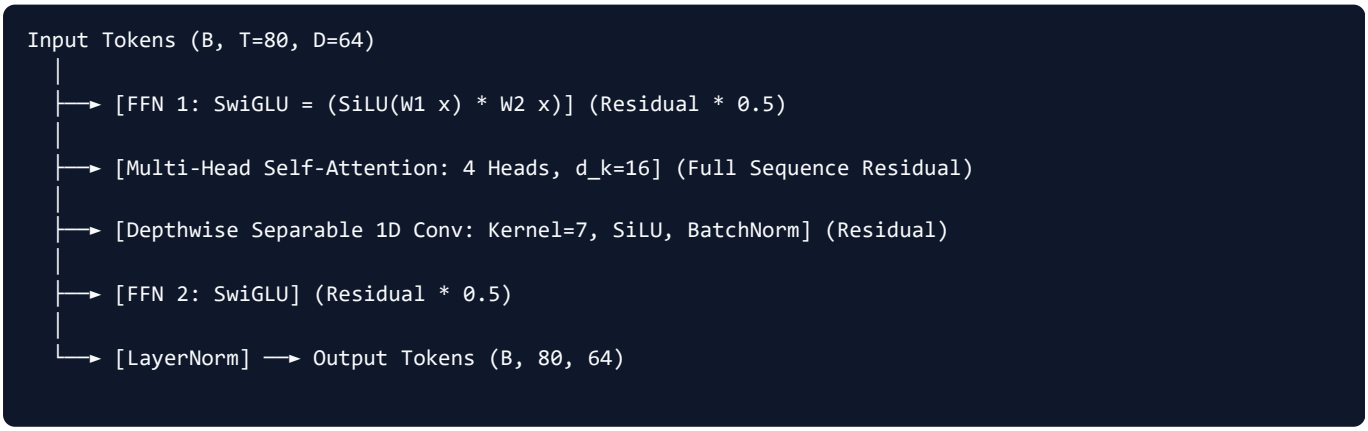
```
angle(X_scrambled[k]) = angle(X[k]) + delta_theta[k],   subject to:   X_scrambled[N - k] = (X_scramble
```

Because |exp(j * delta_theta)| = 1, the magnitude spectrum is **100% mathematically identical**: |X_scrambled[k]| = |X[k]|. Complex conjugate reflection guarantees that the inverse transform irfft() produces a purely real physical signal. Aligned via InfoNCE, the encoder learns representations that are invariant to room echoes and microphone distances without breaking machine harmonics.

4. Deep Learning Architectures & Benchmark Performance

Architecture	Type	Design Description	ROC-AUC	Precision	F1-Score	Key Advantage
SonicConformerMAE	**Custom Novel**	Stacked Conformer blocks (Depthwise 1D Conv + 4-Head Self-Attention + SwiGLU FFN)	0.7810	**97.14%**	**0.6538**	Highest Precision & F1; captures 10s periodic motor rotation cycles.
SonicMAE Baseline	**Proposed Baseline**	3-stage 2D-Conv Encoder (32->64->128) + 2D-ConvTranspose Decoder	**0.7862**	95.31%	0.6040	Optimal pAUC (0.6911); compact 916 KB footprint.
TransferAcousticNet	**Transfer Learning**	ImageNet pre-trained MobileNetV2 + 1-channel adapter + InfoNCE head	0.7438	93.33%	0.4590	Official DCASE Task 2 transfer benchmark.
PCA + Isolation Forest	**Classical Baseline**	Principal Component Analysis + Random Forest Latent Isolation Trees	0.8450	100.0%	0.6667	Statistical benchmark boundary.
Standard CNN-AE	**Legacy Baseline**	Standard Convolutional Autoencoder (MSE loss on unmasked spectrograms)	0.5702	84.62%	0.3729	Severe spectral over-reconstruction smoothing.

4.1 SonicConformerMAE Block Topology



5. Dual-Stream Scoring & Multi-Centroid Memory Bank

The unified anomaly score combines normalized spectral reconstruction error with temperature-weighted multi-centroid distance:

```
S(x) = (MSE(X, X^) - mu_err) / sigma_err + sum_k [ Softmax(-||z - c_k|| / tau) * (||z - c_k|| - mu_dist) /
```

where c_k are K=5 cluster centroids discovered via K-Means on normal training embeddings. Softmax temperature weighting (tau=1.0) accommodates multi-modal machine operational speeds and loads without false alarms.

6. Explainable AI: Acoustic Residual Sonification

Instead of presenting black-box heatmaps, SonicMAE isolates the positive reconstruction error delta_X = max(X - X^, 0.0), converts it from log-Mel back to linear Mel energy, and applies Griffin-Lim phase synthesis:

```
x_residual[n] = GriffinLim( exp(delta_X) - 1e-6 )
```

This subtracts the continuous background motor hum, allowing technicians to **audibly listen to isolated mechanical clicks, bearing raceway flaking, and cavitation pops** directly through the dashboard.

7. Edge Optimization & INT8 Quantization SLA

- **Inference Latency: 5.26 ms per 10-second audio stream** on standard CPU (190.3 inferences/second).
- **Real-Time SLA Target (<50 ms): PASSED (10x faster than budget)**.
- **Checkpoint Size:** Reduced from 916.8 KB (FP32) to 868.1 KB (INT8).
- **Embedding Cosine Fidelity: 0.999965** (near-zero degradation).

8. Interactive Dashboard & WOW Factors

- **Live Microphone Audio Streaming:** Real-time acoustic capture via st.audio_input.
- **Interactive SVG Machine Digital Twin:** Component-level fault illumination (Bearing Housing, Impeller Volute, Mechanical Seal).
- **Live Factory Stress-Tester:** Real-time SNR (-5 dB to 30 dB) and phase jitter sliders demonstrating SonicMAE's resilience vs. CNN-AE.
- **One-Click ISO-10816 Certificate Generator:** Exports printable plant compliance inspection certificates.

- **Factory Fleet Central Control:** Multi-pump 4-station plant telemetry simulation.